



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 09

Aim: To study and implement different Disk Scheduling Algorithms such as FCFS, SSTF, SCAN and C-SCAN and calculate the Total Track Movements required for servicing disk requests.

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CODE :-

```
#include <stdio.h>
#include <stdlib.h>

void sort(int arr[], int n)
{
    for(int i=0;i<n-1;i++)
        for(int j=i+1;j<n;j++)
            if(arr[i] > arr[j])
            {
                int t = arr[i];
                arr[i] = arr[j];
                arr[j] = t;
            }
}

void fcfs(int req[], int n, int head)
{
    int total = 0;

    for(int i=0;i<n;i++)
    {
        total += abs(head - req[i]);
        head = req[i];
    }

    printf("\nTOTAL TRACK MOVEMENTS = %d\n", total);
}

void sstf(int req[], int n, int head)
{
    int total = 0, visited[100]={0};

    for(int i=0;i<n;i++)
    {
        int min = 9999, index = -1;

        for(int j=0;j<n;j++)
        {
            if(!visited[j])
            {
                int dist = abs(head - req[j]);
                if(dist < min)
                {
                    min = dist;
                    index = j;
                }
            }
        }
    }
}
```

```
        }
    }
}

visited[index] = 1;
total += min;
head = req[index];
}

printf("\nTOTAL TRACK MOVEMENTS = %d\n", total);
}

void scan(int req[], int n, int head, int size)
{
    int total = 0;
    sort(req, n);

    int i;
    for(i=0;i<n;i++)
        if(head < req[i])
            break;

    // Move Right
    for(int j=i;j<n;j++)
    {
        total += abs(head - req[j]);
        head = req[j];
    }

    total += abs(head - (size-1));
    head = size-1;

    // Move Left
    for(int j=i-1;j>=0;j--)
    {
        total += abs(head - req[j]);
        head = req[j];
    }

    printf("\nTOTAL TRACK MOVEMENTS = %d\n", total);
}

void cscan(int req[], int n, int head, int size)
{
    int total = 0;
    sort(req, n);
}
```

```

int i;
for(i=0;i<n;i++)
    if(head < req[i])
        break;

// Move Right
for(int j=i;j<n;j++)
{
    total += abs(head - req[j]);
    head = req[j];
}

total += abs(head - (size-1));
head = 0;
total += (size-1);

// Again Move Right
for(int j=0;j<i;j++)
{
    total += abs(head - req[j]);
    head = req[j];
}

printf("\nTOTAL TRACK MOVEMENTS =
%d\n", total);
}

int main()
{
    int choice, n, head, size;

    printf("\nDISK SCHEDULING SIMULATOR\n");
    printf("1. FCFS\n2. SSTF\n3. SCAN\n4. C-
SCAN\n");

    printf("Enter Choice: ");
    scanf("%d", &choice);

    printf("Enter number of track requests: ");
    scanf("%d", &n);

    int req[100];

    printf("Enter track request sequence:\n");
    for(int i=0;i<n;i++)
        scanf("%d", &req[i]);

    printf("Enter current head position: ");
    scanf("%d", &head);

```

```

if(choice == 3 || choice == 4)
{
    printf("Enter Disk Size: ");
    scanf("%d", &size);
}

switch(choice)
{
    case 1: fcfs(req, n, head); break;
    case 2: sstf(req, n, head); break;
    case 3: scan(req, n, head, size); break;
    case 4: cscan(req, n, head, size); break;
    default: printf("Invalid Choice");
}

return 0;
}

```

OUTPUT:-

FCFS:

Run	Output
	<pre>DISK SCHEDULING SIMULATOR 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN Enter Choice: 1 Enter number of track requests: 8 Enter track request sequence: 98 183 37 122 14 124 65 67 Enter current head position: 53 TOTAL TRACK MOVEMENTS = 640</pre>

SSTF:

	Output
	<pre>DISK SCHEDULING SIMULATOR 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN Enter Choice: 2 Enter number of track requests: 8 Enter track request sequence: 98 183 37 122 14 124 65 67 Enter current head position: 53 TOTAL TRACK MOVEMENTS = 236</pre>

SCAN:

n	Output
	<pre>DISK SCHEDULING SIMULATOR 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN Enter Choice: 3 Enter number of track requests: 8 Enter track request sequence: 98 183 37 122 14 124 65 67 Enter current head position: 53 Enter Disk Size: 200 TOTAL TRACK MOVEMENTS = 331</pre>

CSCAN:

n	Output
	<pre>DISK SCHEDULING SIMULATOR 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN Enter Choice: 4 Enter number of track requests: 8 Enter track request sequence: 98 183 37 122 14 124 65 67 Enter current head position: 53 Enter Disk Size: 200 TOTAL TRACK MOVEMENTS = 382</pre>